

DIGITAL CREDENTIAL OPERATIONAL TRUST INFRASTRUCTURE

A Neutral Ledger Between Issuer, Holder, and Employer

Conceptual Framework ▪ Version № 1.0

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ABSTRACT

This document proposes a neutral trust infrastructure for the digital credential ecosystem. The infrastructure does not audit credential programs, certify issuers, or evaluate operational significance claims. It performs the single neutral function of recording what issuers commit to, verifies that holders hold what issuers issued, and makes both queryable by employers and any other party. Evaluation is external and sovereign. The infrastructure is the registry, not the judge. External parties, regulators, holders, employers, researchers, RDA practitioners, etc: evaluate commitments against operational reality. The infrastructure enables that evaluation by making the commitment record available. It takes no position on the accuracy of those commitments. The proposed Digital Credential Operational Trust Infrastructure (DCOTI) simply makes operational significance commitments of credentialing programs attributable. Accountability flows from a cryptographically signed neutral trust ledger.

1 The Neutrality Principle

The trust infrastructure does nothing except be. It exists between three parties: (1) issuer, (2) holder, and (3) employer, without preferential service to any. It serves all three by serving none.

This is not a diplomatic compromise. It is the design requirement. An infrastructure that evaluates claims risks losing neutrality to the party whose claims are being evaluated. An infrastructure that certifies claims becomes captured by the parties who benefit from certification. An infrastructure that adjudicates disputes becomes a litigation body. None of these qualify as a trust infrastructure.

The neutral infrastructure does one thing: it produces a record. The record is public, cryptographically signed, and permanent. It records what each party committed to, when they committed to it, and what they held at each moment in time at the moment of commitment. What the parties do with each other, and what any external observer does with the record, is beyond the scope of the infrastructure's function.

The Neutrality Design Requirement

The infrastructure records commitments. It does not evaluate them. It records holding relationships. It does not validate whether the holder deserves the credential. It records what issuers say their credentials do. It does not verify whether what they say is operationally true. The operational truth evaluation is external. The infrastructure makes it possible by making the commitment visible, attributable, and permanent.

1.1 Why Neutrality Produces Stronger Accountability Than Auditing

The possible response to the divergence between signal representations and operational reality is an audit requirement that compels auditing of credentialing programs by independent bodies who certify the accuracy of their claims. This path retains three structural risks that are costly.

- I. Auditing bodies take sides. The moment an audit body issues a negative finding, the issuer has an adversary with institutional authority. Litigation, regulatory challenge, political pressure, and eventual capture follow.
- II. Auditing requires ongoing resource commitment. Who funds the auditors? If issuers fund the auditors, the auditors are captured. If governments fund the auditors, auditing becomes regulatory and subject to all the public bureaucracy, politicization, and jurisdictional fragmentation that regulatory enforcement entails.
- III. Auditing produces binary outcomes where the reality is continuous. A credential program is not either compliant or non-compliant. Forcing this continuous reality into a binary certification creates false precision and hides the specific gaps that matter.

Neutrality produces stronger accountability through a different mechanism where the issuer, in registering their credential program, must make their operational significance commitments explicit, machine-readable, cryptographically attributed, and permanently queryable. They are not being evaluated. They are being recorded, and that act, of being recorded, changes the nature of their commitment: publicly, attributably, and permanently.

6 The AI Complication

6.1 AI as Credential Issuer

AI-assessed credentials, where competency determination is made by an AI system rather than a human evaluator, create a new version of the commitment field problem. The issuer's commitment now includes a claim about what the AI assessment system does: how it evaluates competency, what inputs it uses, what its accuracy profile is, and how its outputs correspond to the competency it claims to measure.

This produces an AI-specific complication in which the AI system's behavior is constantly evolving as the model is continuously updated. An AI assessment system that accurately measured a competency in 2024 may measure a different thing in 2026 if the underlying model has changed. The DCOTI's temporal commitment field structure is specifically designed for this: issuers must specify the model version or version range for which their commitment is valid, and must update or expire commitments when the underlying model changes materially.

6.2 AI as Employment Evaluator

When the employer's evaluation system is itself an AI, as is increasingly the case in ATS deployment, the question of what happens to a credential at the employer-side system boundary becomes more complex. A human evaluator who reads a credential on a resume makes a judgment. An AI ATS that processes an application through a classification model may not have a mechanism for receiving or weighting a credential at all.

The DCOTI does not resolve this complexity. It records the issuer's commitment. What the employer's AI system does with that commitment is within the employer's own operational architecture. The practical implication is, as AI ATS systems become universal, the DCOTI's standardized commitment fields can serve as the data source for employer-side AI systems that are designed to process them, but only where the employer chooses to build that integration.

The Neutrality of AI Integration

The DCOTI does not require employers to build AI integrations with credential commitment data. It records what each issuer commits their platform does. If an issuer commits to ATS integration and the employer's AI system receives no credential-linked signal, an external party using RDA can document that gap. The infrastructure made the commitment attributable. It did not police the integration. External evaluation is concerned with responding to the gap.

Publication and Prior Art Notice

This document constitutes an original intellectual contribution establishing the conceptual framework for the Digital Credential Operational Trust Infrastructure (DCOTI). The neutral ledger design principle, three-party architecture, commitment field taxonomy, enrollment-time snapshot mechanism, and governance design described herein are original contributions of the author. This document is issued as a companion to the Runtime Divergence Analysis methodology specification (Version 1.0, May 2026) and its Supplement on Distributed System and AI Application. DCOTI is the infrastructure within which RDA findings have their highest systemic utility, and RDA is the external evaluation methodology best suited to assessing the gap between DCOTI-registered commitments and operational reality. Author: Mohamed S. Mohamed • May 2026

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